

9front Cybersecurity Hardening Guide — Community Edition

Anastasios Papalias

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AnastasiosPapalias/Plan9](https://github.com/AnastasiosPapalias/Plan9)

This is the free community edition. The Full Edition
(200 pages) is available via Acopon Digital Publica-
tions.

Cover: visual concept developed with generative tools.

Chapter 1: The Philosophy of Security in Plan 9

Plan 9 security is architectural, not additive. Security
properties emerge from the system design —
namespace isolation, uniform file interface, per-process
resource mapping — not from layers applied over an
existing base.

**SURFACE: FOUNDA-
TION**

Resource access

SURFACE: FOUNDATION

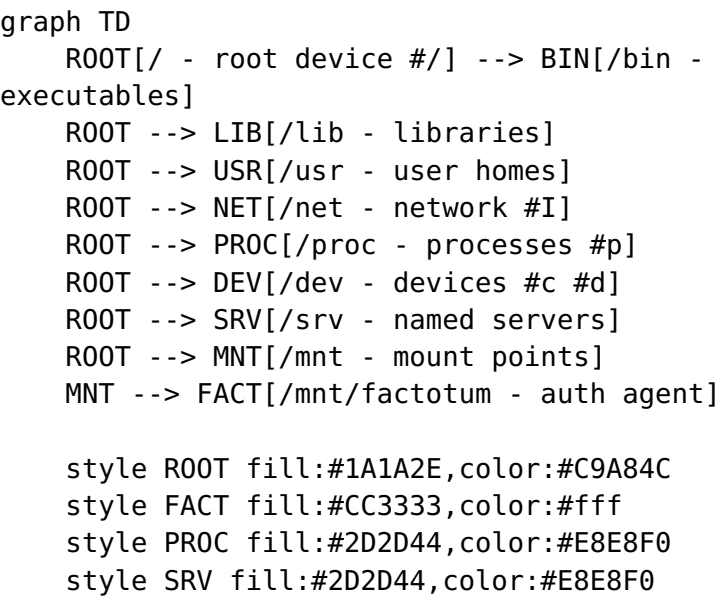
	Name in namespace → 9P server → file operation
No-access semantics	Name absent from namespace = resource does not exist
Stronger than ACL	Cannot bypass a missing namespace entry; can by- pass a permission check
Trust anchor	Trust granted to servers, not to filesystem layer

Saltzer & Schroeder: Eight Principles [7]

Principle	Plan 9 Implementation
Economy of mechanism	38 syscalls; 5MB kernel; one file protocol; all se- curity statable in one paragraph
Fail-safe defaults	rfork(RFCNAMEG) starts process with empty namespace. No inherited access.
Complete mediation	Every file operation is a 9P message processed by the server. No bypass.
Open design	9front source publicly available. Security not predicated on secrecy.
Separation of privilege	Auth requires user key (factotum) AND valid re- sponse from authsrv.
Least privilege	RFCNAMEG enables ser- vices with only the re- sources they need.
Least common mechan- ism	Per-process namespaces eliminate shared filesys- tem state across trust levels.
Psychological acceptabil- ity	Weakest: namespace model non-intuitive. EN- OENT not EPERM. Mis- configuration risk.

Chapter 2: Namespaces as Isolation

Figure: Namespace Tree



A Plan 9 namespace is a per-process mapping from path names to 9P servers. It is not a restriction on a global view — it is the entirety of what the process can see.

Construction Syscalls

Operation	Effect
bind(src, dst, MRE-PLACE)	Replace existing bindings at dst with src
bind(src, dst, MBEFORE)	Add src BEFORE existing bindings (first in union resolution)
bind(src, dst, MAFTER)	Add src AFTER existing bindings (last in union resolution)
mount(fd, afd, dst, flags, aname)	Attach 9P server on fd to dst; afd=-1 = unauthenticated (local only)

rfork Isolation Flags

Flag	Effect	Security Use
RFNAMEG	Copy of parent namespace	Standard service isolation
RFCNAMEG	EMPTY namespace	Maximum isolation — PRE-FERRED for new services
RFFDG	Copy of fd table	Prevents fd leakage
RFENVG	Copy of /env	Prevents env injection
RFNOTEG	New note group	Prevents note injection from parent
RFMEM	SHARES parent memory	NEVER for service children

Attack Vectors

SURFACE: NAMESPACE	
9f-001 MBEFORE injection	bind attacker dir over /bin MBEFORE; parent resolves attacker first; requires shared namespace
9f-010 Rogue /srv	Register attacker 9P server in /srv before legitimate server; victim mounts by name
9f-011 NS inheritance	rfork without RFNAMEG; compromised child binds into shared namespace; ATT&CK T1611 (analogical)
Recon	/proc/n/ns is owner-readable; reveals all bind/mount entries and ordering

Correct flag set for network service child processes:
`rfork(RFPROC|RFNAMEG|RFFDG|RFENVG)`

Chapter 3: Files and Trust Boundaries

Every resource in Plan 9 is served through 9P. The uniformity means one interface to secure. The cost: trust is placed in servers, not in a kernel enforcement layer.

The /srv Directory

SURFACE: FILESYSTEM

Purpose	Kernel-maintained registry of named 9P server file descriptors
Registration	Server creates /srv/; writes file descriptor to it
Mount client	Client reads /srv/ to obtain fd; then calls mount(fd, path, flags)
Attack risk	Name collision: attacker registers /srv/kfs before legitimate server starts
Permission	Standard Plan 9 file permissions (uid/gid/other read/write)

Attach Boundary

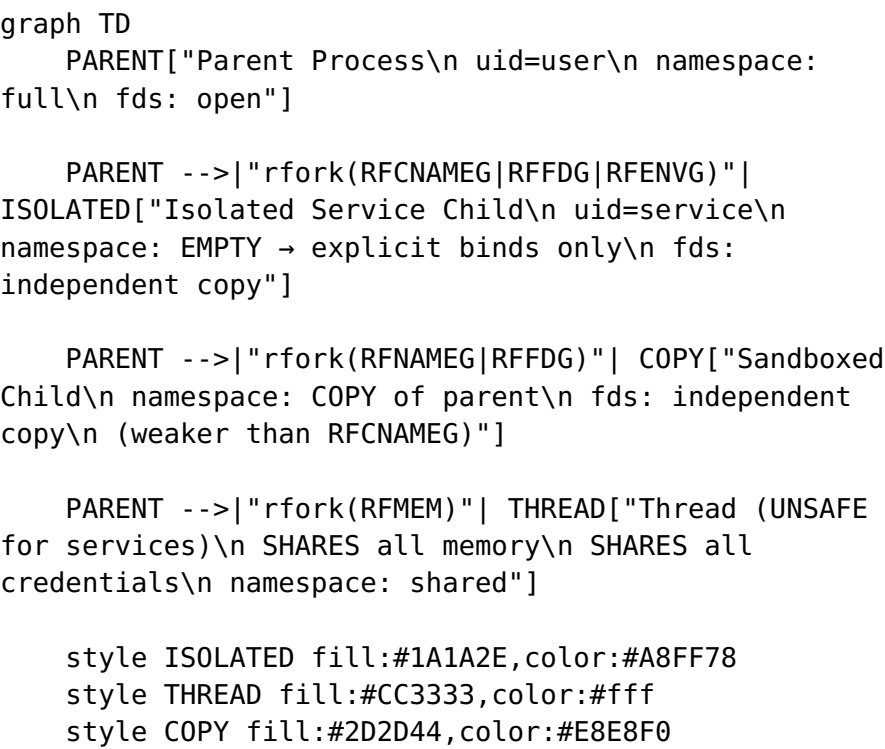
Method	Security
Tauth + Tattach (authenticated)	factotum conducts 3-party exchange before access granted
Tattach afid=NOFID (unauthenticated)	No identity established; server MUST reject from remote clients
Local pipe (unauthenticated)	Acceptable — identity established by process context
Network TCP (unauthenticated)	OPEN DOOR to entire exported namespace — never acceptable

Evaluation Checklist

For each mounted server: what is its provenance? Who started it? What uid does it run as? For each /srv entry: was it created at startup? Does it still point to the expected server? For each network export: is TLS active? Is authenticated attach required?

Chapter 4: Process Model and rfork

Figure: Process Model and rfork



Processes are files. /proc exposes every running process as a directory. Security of process isolation is security of /proc access.

/proc File Interface

File	Permissions	Contents	Security Risk
/proc/n/status	World-readable	PID, name, uid, gid, state, memory	Reconnaissance

File	Permissions	Contents	Security Risk
/proc/n/ns	Owner-readable	Full namespace map	Complete recon target — capture first in IR
/proc/n/mem	Owner/hostowner RW	Raw address space	CRITICAL: key theft (9f-008)
/proc/n/text	Owner-readable	Executable image	Integrity verification; identity check
/proc/n/ctl	Owner write-only	kill, stop, start, profile	DoS via kill
/proc/n/note	Owner write-only	Deliver note (signal)	Note injection (9f-009)
/proc/n/fd	Owner-readable	Open fd inventory	Credential leakage (9f-012)

SURFACE: /PROC

Uid boundary	Only isolation boundary within /proc. Same uid = full access to mem, ctl, note.
Hostowner	Reads any process mem regardless of uid.
Service design	Dedicated uid per service limits /proc blast radius.
9f-008 mem read	Same-uid read of /proc/target/mem extracts session keys, factotum keys. ATT&CK T1055.
9f-009 note inj.	Write unexpected note to /proc/target/note. ATT&CK T1055.008.
9f-012 fd leak	Read /proc/target/fd to enumerate open resources. ATT&CK T1083.

Chapter 5: 9P and Network Exposure

Figure: 9P Protocol Message Flow

```
sequenceDiagram
    participant C as Client
    participant S as 9P Server
    participant A as AuthSrv

    C->>S: Tversion (negotiate msize, version)
    S->>C: Rversion

    C->>S: Tauth (request auth fid)
    S->>C: Rauth (afid established)

    Note over C,A: Three-party auth exchange via afid
    C->>A: factotum rpc (p9sk1 / dp9ik)
    A->>C: session ticket

    C->>S: Tattach (fid=1, afid=authenticated)
    S->>C: Rattach (root qid)

    C->>S: Twalk (fid=1, newfid=2, path=bin/rc)
    S->>C: Rwalk (qids)

    C->>S: Topen (fid=2, mode=OREAD)
    S->>C: Ropen (qid, iounit)

    C->>S: Tread (fid=2, offset=0, count=512)
    S->>C: Rread (data)

    C->>S: Tclunk (fid=2)
    S->>C: Rclunk
```

9P is the universal resource access protocol. Every local file operation is 9P internally. Every network file export is 9P explicitly.

Message Header

size[4] type[1] tag[2] size: total message bytes including size field type: T/R-message type (Tversion=100, Tattach=104, ...) tag: transaction identifier; NOTAG=0xFFFF for Tversion only

Key Protocol Elements

Element	Definition
fid	32-bit handle for an open file reference in a session. NOFID=0xFFFFFFFF.
qid	13-byte unique file identifier: type[1] + version[4] + path[8].
Tversion	First message. Negotiates max message size (msize). Must use NOT-AG.
Tattach	Establish session root. afid=NOFID = unauthenticated (reject from network).
Twalk	Traverse path components from existing fid. Max 16 components per message.
Topen	Open a fid for I/O. Returns qid and iounit.
SURFACE: 9P PROTOCOL	
9f-005 unauth attach	Tattach afid=NOFID accepted. Full namespace access. ATT&CK T1133. CRITICAL.
9f-006 fid exhaustion	Open fids without Tclunk fills fid table. ATT&CK T1499. Mitigation: per-session limit <=32.
9f-007 replay	Capture auth Tattach; replay to different server. ATT&CK T1557. Mitigation: TLS + dp9ik nonces.
Path traversal	Walk '..' components to escape export root. Mitigation: exportfs -r tracks position.

Required Network Controls

```
aux/listen1 tcp!*!564 /bin/tlssrv -a /bin/exportfs -  
r /exported/path # TLS required for all network  
9P — no exceptions # Reject Tattach  
afid=NOFID from non-loopback addresses #  
Per-session fid limit (<=32 recommended) #  
Port 564: IANA-assigned for plan9file
```

Chapter 6: Factotum and Authentication

Figure: Factotum Authentication Flow

sequenceDiagram

```
participant CF as Client Factotum  
participant AS as AuthSrv  
participant SF as Service Factotum
```

Note over CF,SF: Phase 1 - Contact AuthSrv

CF->>AS: identity request (uname, sname)

AS->>AS: generate K_session

AS->>CF: ticket_C (encrypted with K_ca)

AS->>SF: ticket_S (encrypted with K_sa)

Note over CF,SF: Phase 2 - Service Exchange

SF->>CF: forward ticket_C

CF->>CF: decrypt ticket_C with K_ca

CF->>SF: ticket_S + authenticator(K_session)

Note over CF,SF: Phase 3 - Mutual Verification

SF->>SF: decrypt ticket_S with K_sa

SF->>SF: verify authenticator

SF->>CF: Rattach (session established)

Note over CF,SF: K_ca and K_sa never leave their owners

Factotum is the authentication agent. It holds credentials and performs authentication on behalf of all processes. Applications never handle credentials directly.

Interface

SURFACE: FACTOTUM

Mount point	/mnt/factotum (userspace 9P server, mounted in process namespace)
rpc file	Authentication channel. Write start message; read/write protocol exchange.
ctl file	Key management. Add, delete, list keys. Confidential attributes prefixed with '!'. Block until factotum needs user input for interactive key entry.
needkey	

Authentication Protocols

Protocol	Description	Source
p9any	Negotiation wrapper. Advertises supported protocols; vulnerable to downgrade without TLS.	Bell Labs
p9sk1	Shared-secret challenge-response. Three-party: client factotum + authsrv + service key.	auth.pdf [4]
dp9ik	9front-specific Diffie-Hellman protocol. Forward secrecy. NOT in Bell Labs auth.pdf.	sys/src/cmd/auth/ (9front only)
secstore	Encrypted persistent key storage. PAK exchange —	9front FQA [6]

Protocol	Description	Source
	passphrase never transmitted.	
SURFACE: FACTOTUM		
9f-003 RPC intercept	Same-uid process writes to /mnt/factotum/rpc. ATT&CK T1552. CRITICAL.	
9f-004 key theft	Same-uid reads plaintext key file before factotum loads. Eliminated by secstore.	
Rogue mount	Attacker binds different server over /mnt/factotum. ATT&CK T1574.	
Protocol downgrade	Strip strong options from p9any. Requires network position + no TLS.	
Memory extraction	Same-uid reads /proc/factotum/mem. ATT&CK T1055. Mitigation: dedicated uid.	

Chapter 7: Debugging and Observation

Debugging and attacking use the same interface: /proc. Access control (uid-based file permissions) is the only separation.

System Observation Files

File	Access	Contents
/dev/sysstat	World-readable	Per-CPU: scheduled procs, context switches, syscalls, faults
/dev/swap	World-readable	Physical memory, free memory, swap

File	Access	Contents
		usage, kernel memory
/proc/n/ns	Owner-readable	Full namespace map — complete trust topology
/proc/n/mem	Owner/hostowner	Raw address space — all in-memory secrets

What Polling Can and Cannot Detect

Can Detect	Cannot Reliably Detect
Persistent namespace modifications (bind/mount that survives poll interval)	Transient processes (start, act, exit within one poll interval)
New /srv registrations	Individual system call events (no audit bus equivalent)
New running processes	File access events (no inotify equivalent)
Changed executable hashes (/proc/n/text)	Packet-level network events (no pcap in base system)
Unexpected network connections (/net/tcp/*/remote)	In-memory key extraction (leaves no filesystem trace)

Monitoring Cadence

Observable	Interval	Rationale
Namespace + /srv	30-60 seconds	Exposure window should be measured in seconds
Executable hashes	5-10 minutes	Binary changes are not instantaneous
Process list	60 seconds	Cross-reference parent pid against known tree
	60 seconds	

Observable	Interval	Rationale
Network connections		Alert on connections to 9P ports from unexpected sources

Chapter 8: Threat Modeling

Figure: Attack Surface Map

```
graph LR
    NET["Network\n(TCP)"] -->|Tattach afid=NOFID|
    9P["9P Server\n9f-005 CRITICAL\n9f-006 MEDIUM\n9f-007 HIGH"]

    LOCALUSER["Local User\n(same uid)"] -->|bind MBEFORE|
    NS["Namespace\n9f-001 HIGH\n9f-002 HIGH\n9f-010 HIGH\n9f-011 HIGH"]

    LOCALUSER -->|/proc/n/mem read|
    PROC["/proc\n9f-008 CRITICAL\n9f-009 MEDIUM\n9f-012 HIGH"]

    LOCALUSER -->|/mnt/factotum/rpc|
    FACT["Factotum\n9f-003 CRITICAL\n9f-004 CRITICAL"]

    style 9P fill:#1A1A2E,color:#C9A84C
    style NS fill:#1A1A2E,color:#C9A84C
    style PROC fill:#1A1A2E,color:#C9A84C
    style FACT fill:#CC3333,color:#fff
```

Four primary attack surfaces. Every significant 9front threat interacts with at least one.

Threat Dataset Summary

ID	Name	Surface	ATT&CK	Severity
9f-001	namespace_escape	namespace	T1574	HIGH
9f-002	namespace_injection	namespace	T1574	HIGH
9f-003	factotum_rpc_interception	factotum	T1552	CRITICAL
9f-004	factotum_key_theft	factotum	T1552.004	CRITICAL

ID	Name	Surface	ATT&CK	Severity
9f-005	9p_unauthenticated_attach	9p	T1133	CRITICAL
9f-006	9p_fid_exhaustion	9p	T1499	MEDIUM
9f-007	9p_replay_attack	9p	T1557	HIGH
9f-008	proc_memory_read	/proc	T1055	CRITICAL
9f-009	proc_note_injection	/proc	T1055.008	MEDIUM
9f-010	srv_rogue_registration	namespace	T1574	HIGH
9f-011	rfork_namespace_inheritance	namespace	T1611*	HIGH
9f-012	fd_leakage	/proc	T1083	HIGH

- T1611 (Escape to Host) is an analogical mapping for rfork namespace inheritance. No Plan 9-specific ATT&CK technique exists.

STRIDE by Surface

Surface	Spoofing	Tampering	Info Disc.	DoS	EoP
namespace	Rogue /srv server	bind injection	ns enumeration	mount loop	bind over / bin
factotum	Rogue factotum mount	key modification	RPC interception	RPC exhaustion	proto downgrade
9P network	Server impersonation	message replay	plaintext capture	fid exhaustion	unauth attach
/proc	Process impersonation	ctl injection	mem/fd read	kill flood	note injection

Chapter 9: Hardening Strategies

Hardening is the systematic reduction of attack surface through deliberate namespace construction, correct rfork flag use, authenticated network exposure, and credential protection.

SURFACE: ALL SUR- FACES

Namespace start	All privileged services start with rfork(RFCNAMEG) — empty namespace
Bind discipline	Bind exactly the directories needed. Document the sequence. Audit against documentation.
Union ordering	More trusted server bound MBE-FORE less trusted.
Factotum scope	Do not include /mnt/factotum in namespaces of services that do not authenticate.
rfork flags	rfork(RFPROC RFNAMEG RFFDG RFENVG) for all network service children
NEVER RFMEM	RFMEM for service children shares all memory including secrets. Never use.
9P TLS	TLS required for all network 9P. tlssrv handles wrapping.
9P auth	Reject Tattach afid=NOFID from

**SURFACE:
ALL SUR-
FACES**

	non-loopback ad- dresses.
fid limit	Reject Twalk when session fid count exceeds threshold (≤ 32 recommen- ded).
Secstore	auth/secstore for all persistent key storage. No plain- text key files on disk.
Uid isola- tion	Dedicated uid per service. Limits / proc-based lateral movement.

**# Required configuration for any network-ex-
posed 9P server: `aux/listen1 tcp!*!564 /bin/tlssrv
-a /bin/exportfs -r /exported/path`**

OpenBSD Comparison

Approach	Mechanism	When Ex- cluded
Plan 9 RFCNAMEG + bind	Construction- time restriction. Excluded paths never existed. Cannot be by- passed.	Not retrofittable — requires design from scratch
OpenBSD pledge(2)	Runtime syscall restriction. SIG- ABRT on viola- tion; ENOSYS with 'error' promise.	Retrofittable to existing pro- grams
OpenBSD un- veil(2)	Runtime filesys- tem restriction. Unveiled paths visible; others return ENOENT.	Retrofittable but weaker than construction- time

Chapter 10: Detection Engineering

Detection in Plan 9 is filesystem polling. No kernel audit framework. Observable surface: /proc, /dev/sysstat, /dev/swap, /srv, /net.

Detection Capability 1: Namespace Integrity

```
#!/bin/rc # baseline:
sha1sum /proc//ns > /
etc/security/ns-baseline
fn check_ns { for pid in /
proc/^/ns { n={echo
$pid\|sed 's\|/proc/\|
\|;s\|/ns\|\|'|}
h={sha1sum $pid 2>/dev/
null|awk '{print $1}'}
if(! ~ $h '' && ! grep -q
h/etc/security/ns -
whitelist2 > /dev/
null)echoNS_A LERTpid=n
hash=$h `{date} } } #
D3FEND: D3-PM (Pro-
cess Monitoring)
```

Detection Capability 2: Executable Integrity

```
fn check_text { for pid in /
proc/^*/text { n={echo
$pid\|sed 's\|/proc/\|\|;s\|/
text\|\|'|}
h={sha1sum $pid 2>/dev/
null|awk '{print $1}'} if(!
~ $h '' && ! grep -q h/etc/
security/text - whitelist2
> /dev/
null)echoTEXT_A LERTpid=n
hash=$h `{date} } } #
D3FEND: D3-SFV (Soft-
ware File Verification)
```

Detection Capabilities 3-5

Capability	Observable	D3FEND	Alert Condition
3: /srv monitoring	ls -la /srv diff	D3-BA (Baseline)	Any new or removed / srv entry
4: Network connections	/net/tcp/*/remote	D3-NTA (Network Traffic Analysis)	Address not in whitelist
5: Process audit	/proc/n/status	D3-PM (Process Monitoring)	Unknown parent pid

Alert delivery: Write alerts to remote syslog destination — not local log file (attackers can modify local logs).

Chapter 11: Incident Response

Incident response on 9front is namespace forensics. Capture before remediation. Scope by uid of compromised process.

Phase 1: Capture

```
#!/bin/rc stamp={date\|
tr ' /' '__'} capdir=/
tmp/ir-$stamp mkdir -p
$capdir/ns $capdir/status
$capdir/fd $capdir/text
for pid in{ls /proc}
{ if(test -d /proc/$pid)
{ cp /proc/$pid/ns
capdir/ns/pid 2>/dev/
null cp /proc/$pid/
status capdir/status/
pid 2>/dev/null cp /
proc/$pid/fd capdir/fd/
pid 2>/dev/null
sha1sum /proc/$pid/
text > capdir/text/pid
2>/dev/null } } ls -la /
srv > $capdir/srv cat /
net/tcp/*/remote >
$capdir/tcp 2>/dev/null
```

Phases 2-4

Phase	Action	Command
2: Ana- lysis	Namespace diff against baseline; unknown ex- ecutables	diff capdir/ns/ * baseline; sha1sumcheck 3 : Containment Removerogue/ srv;killcompromisedprocess rm/srv/suspicious; echokill > /proc/pid/ctl
4: Re- covery	Key rota- tion; sec- store update	auth/changeuser; kill factot- um; restart factotum

Chapter 12: AI-Driven Security

The companion security_dataset.jsonl encodes the threat analysis in STIX 2.1 format, enabling AI systems to reason over 9front security knowledge directly.

Dataset Schema

SURFACE: AI LAYER	
	NDJSON
Format	— one STIX 2.1 attack-pattern object per line
STIX type	attack-pattern with spec_version 2.1
ID pattern	attack-pattern-9front-NNN
Extension	x_9front_prefix per STIX 2.1 custom property convention
x_9front_vector	Specific mechanism — the exact operation an attacker performs
x_9-front_surface	Attack surface: namespace 9p /proc factotum
x_9-front_detection	What to poll; what constitutes an alert condition
x_9-front_mitigation	Specific system calls and file paths for remediation

**SURFACE: AI
LAYER**

x_9- front_severity	CRITICAL HIGH MEDIUM
x_9- front_references	Primary source documents for tech- nical claims

ATT&CK Mapping Quality

Quality	Techniques	Note
Direct	T1133, T1499, T1552, T1552.004, T1557, T1083, T1055	Direct enter- prise equival- ents exist
Parent pre- ferred	T1574	Use parent, NOT T1574.006 (dynamic linker — wrong mech- anism)
Analogical	T1611	Escape to Host — analogical for rfork namespace in- heritance; al- ways flagged

NIST AI RMF Alignment [9]

Function	Implementation
GOVERN	Dataset limitations docu- mented: no Plan 9 ATT&CK; all mappings analogical; dp9ik source noted.
MAP	12 entries cover all four primary attack surfaces.
MEASURE	Technical claims backed by primary sources [1]- [6].
MANAGE	

Function	Implementation
	Dataset versioned on GitHub; updates as threat landscape evolves.

NIST AI RMF is a voluntary framework. It recommends practices; it does not mandate them.

Threat Dataset

(security_dataset.jsonl)

— Summary

ID	Name	Surface	ATT&CK	Severity	Key Mit
9f-001	namespace_escape	namespace	T1574	HIGH	rfork(RFpoll /pro
9f-002	namespace_injection	namespace	T1574	HIGH	Monitorfy ownerfore mou
9f-003	factotum_rpc_intercept	factotum	T1552	CRITICAL	Dedicateclude /mum fromns
9f-004	factotum_key_theft	factotum	T1552.004	CRITICAL	auth/secno plainon disk
9f-005	9p_unauth_attach	9p	T1133	CRITICAL	tlssrv -a,afid=NCnetwork
9f-006	9p_fid_exhaustion	9p	T1499	MEDIUM	Per-sesslimit <=
9f-007	9p_replay_attack	9p	T1557	HIGH	TLS + dsion non
9f-008	proc_memory_read	/proc	T1055	CRITICAL	Dedicate service
9f-009	proc_note_injection	/proc	T1055.008	MEDIUM	RFNOTEfork; decauid
9f-010	srv_rogue_registration	namespace	T1574	HIGH	Monitoralert on change
9f-011	rfork_ns_inheritance	namespace	T1611*	HIGH	

ID	Name	Surface	ATT&CK	Sever-ity	Key Mit
9f-012	fd_leakage	/proc	T1083	HIGH	rfork(RF always f children RFCFDC close un before e

- T1611 is an analogical mapping. T1574.006 is never used (dynamic linker hijacking — wrong mechanism for Plan 9 bind misuse). All ATT&CK mappings are analogical; ATT&CK contains no Plan 9-specific techniques.

Bibliography & Sources

All technical claims in this edition trace to primary sources. The 9front FQA at fqa.9front.org is the authoritative reference for 9front-specific behavior.

Ref	Source	URL
[1]	Plan 9 System Overview	doc.cat-v.org/.../plan9.pdf
[2]	Namespaces in Plan 9	doc.cat-v.org/.../namespaces.pdf
[3]	9P Protocol	doc.cat-v.org/.../9p.pdf
[4]	Authentication (Bell Labs) — p9sk1 ONLY	doc.cat-v.org/.../auth.pdf
[5]	Plan 9 Security	doc.cat-v.org/.../security.pdf
[6]	9front FQA (authoritative for dp9ik)	fqa.9front.org/fqa.pdf
[7]	Saltzer & Schroeder 1975 — Proc. IEEE Vol.63 No.9 pp. 1278-1308	web.mit.edu/.../protection.pdf
[8]	NIST SP 800-30 Rev 1	nvlpubs.nist.gov
[9]		

Ref	Source	URL
	NIST AI RMF (AI 100-1) — Voluntary	nvlpubs.nist.gov/.../AI.100-1.pdf
[10]	ATT&CK Enterprise (all Plan 9 mappings analogical)	attack.mitre.org
[11]	STIX 2.1 Specification	docs.oasis-open.org/.../stix-v2.1
[12]	D3FEND	d3fend.mitre.org
[13]	OpenBSD pledge(2)	man.openbsd.org/pledge.2
[14]	OpenBSD unveil(2)	man.openbsd.org/unveil.2
[15]	Anderson: Security Engineering 3e	cl.cam.ac.uk/~rja14/Papers/SEv3.pdf

Note on dp9ik: 9front-specific. Not documented in Bell Labs auth.pdf [4]. Source: sys/src/cmd/auth/ in 9front repository. Note on ATT&CK mappings: All mappings are analogical. ATT&CK contains no Plan 9-specific techniques.

Companion Files

File	Description
security_dataset.jsonl	12 STIX 2.1 attack-pattern objects. One per line. Machine-readable threat dataset.
README.md	GitHub landing page. Overview, usage, license, technical accuracy notes.
diagrams/namespace_tree.md	Mermaid: Plan 9 namespace hierarchy diagram.
diagrams/9p_flow.md	Mermaid: 9P protocol message flow sequence diagram.
diagrams/process_model.md	Mermaid: rfork isolation model diagram.

File	Description
diagrams/factotum_auth.md	Mermaid: Three-party factotum authentication sequence.
diagrams/attack_surface_map.md	Mermaid: Complete attack surface map.

Dataset Field Reference

```
{ "id": "attack-pattern-9front-NNN",
  "type": "attack-pattern", "spec_version": "2.1",
  "name": "",
  "kill_chain_phases":
  [{ "kill_chain_name":
    "mitre-attack",
    "phase_name":
    "..."}], "external_references":
  [{ "source_name":
    "mitre-attack", "external_id": "T...",
    "description": "mapping quality note"}],
  "x_9front_vector":
  "specific attack operation", "x_9front_surface":
  "namespace | 9p | /proc | factotum",
  "x_9front_detection":
  "what to poll, alert condition", "x_9front_mitigation":
  "specific syscalls and paths", "x_9front_severity":
  "CRITICAL | HIGH | MEDIUM", "x_9front_references":
  ["source.pdf"] }
```